# MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

**COLLEGE OF INFOMATION AND COMMUNICATION TECHNOLOGY**

**MODULE NAME;** MOBILE APPLICATION DEVELOPMENT.

**MODULE CODE;** 8203

**FACILITATOR;** RUKOIJO A.

**MODE OF ASSIGNMENT;** GROUP PROJECT

**ACADEMIC YEAR;** 2025/2026.

**SEMISTER;** ONE.

PROJECT TITLE; ON-DEMAND CLEANING AND SANITATION SERVICES PLATFORM

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# SafishaNasi APP

## 1. Introduction

### 1.1 Background

With the increasing demand for cleaning services in homes, offices, and institutions, many people face difficulties finding reliable cleaners quickly. Most traditional methods rely on word of mouth, local contacts, or informal arrangements, which are often inefficient and unreliable.

The SafishaNasi App was developed to solve this problem by providing an online platform where customers can easily find and book professional cleaning services. The application connects customers with cleaners and allows bookings to be managed digitally through a simple web interface.

The system also helps cleaners receive job requests and enables administrators to manage the platform effectively.

### 1.2 Problem Statement

Many households, offices, and schools struggle to find trusted cleaning services efficiently. Traditional systems lack digital booking, centralized service management, and proper communication between customers and cleaners.

The SafishaNasi App addresses this issue by providing a modern web-based platform that simplifies the process of booking cleaning services.

### 1.3 Project Objectives

• To develop a web-based platform for booking cleaning services.
• To allow customers to easily view available cleaning services.
• To enable customers to book cleaning services online.
• To provide cleaners with a system to receive job requests.
• To allow administrators to manage services and monitor bookings.

## 2. System Requirements

### 2.1 Functional Requirements

User Registration – Customers and cleaners must be able to create accounts in the system.

User Login – Users must log into the system using their email and password.

View Services – Customers must be able to view available cleaning services such as:
• Home Cleaning

• Office Cleaning
• School / College Cleaning

Booking Services – Customers must be able to book a cleaning service through the platform.

Cleaner Dashboard – Cleaners must be able to view available cleaning jobs assigned to them.

Admin Management – The admin must be able to manage services and monitor system activities.

CRUD Operations – The system must allow Create, Read, Update, and Delete operations for services and bookings.

**2.2 Non Functional Requirements**
Usability – The system must be easy to use and understandable for all users.

Performance – The application must respond quickly and handle multiple users.

Security – User data must be protected using authentication mechanisms.

Responsiveness – The system must work well on both mobile devices and desktop computers.

Maintainability – The code must be well organized and easy to maintain.

## 3. System Architecture
The SafishaNasi App follows a client-server architecture consisting of three main components.

**Frontend**
The frontend displays the user interface and communicates with the backend using API requests.
Technologies used include HTML, CSS, JavaScript, and Font Awesome icons.

**Backend**
The backend processes requests, handles business logic, and communicates with the database.
Technologies used include Node.js and Express.js.

**Database**
The database stores application data such as users, services, and bookings.
Technology used: MySQL database managed through phpMyAdmin.

## 4. Project Folder Structure
The project is organized into two main sections: frontend and backend. This structure helps maintain a clean and organized project.

**5. Database Design**
Users Table – stores user account information (id, name, email, password, role).

Services Table – stores available cleaning services (id, service_name, description, price).

Bookings Table – stores customer bookings (id, user_id, service_id, booking_date, status).

Reviews Table – stores feedback from customers.

Payments Table – stores payment information related to bookings.

**6. Security Considerations**
Authentication – Users must log in using credentials before accessing system services.

Password Protection – Passwords are stored securely using hashing techniques.

Token-Based Authentication – Tokens maintain secure communication between frontend and backend.

Role-Based Access Control – Different user roles have different permissions (Customer, Cleaner, Admin).

**7. Implementation**
The SafishaNasi App was implemented using a full stack web development approach.

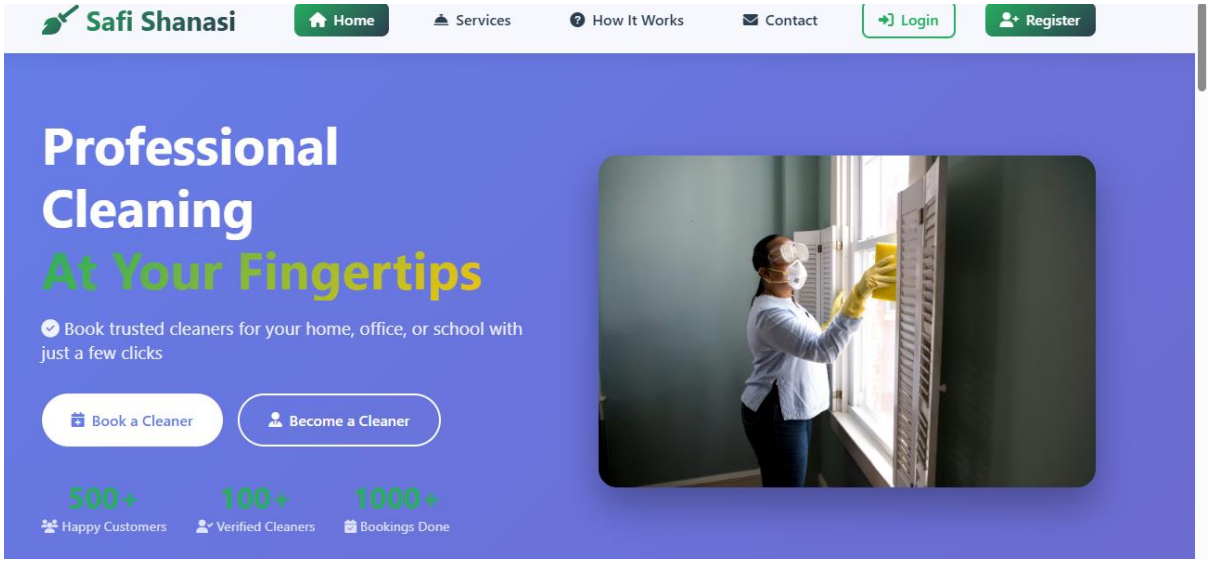
Frontend Implementation – HTML, CSS, and JavaScript were used to build interactive user interfaces.

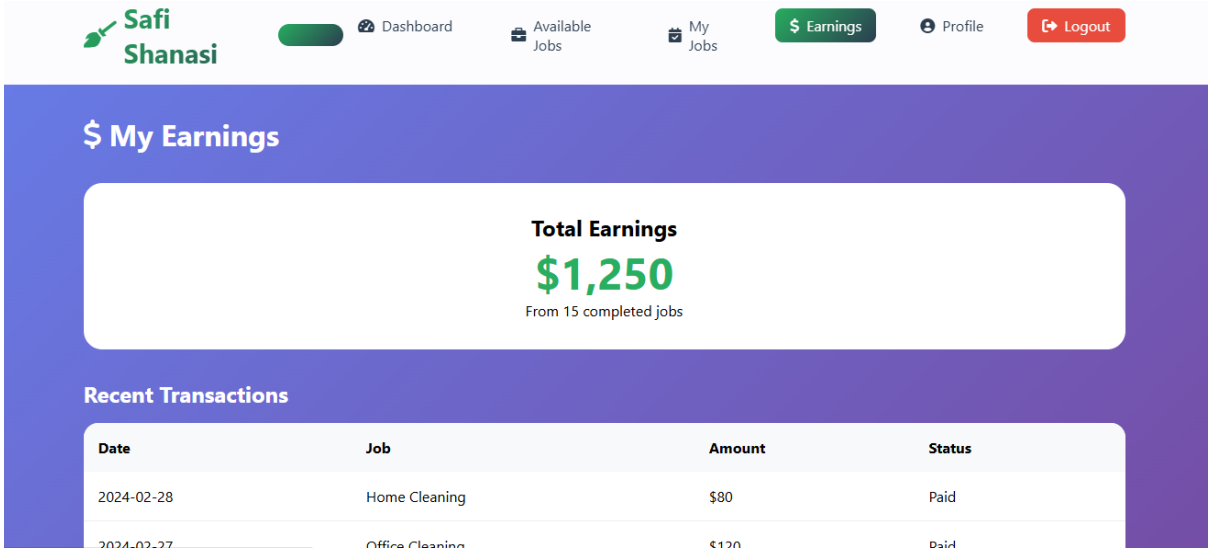
Backend Implementation – Node.js and Express.js were used to build API endpoints.

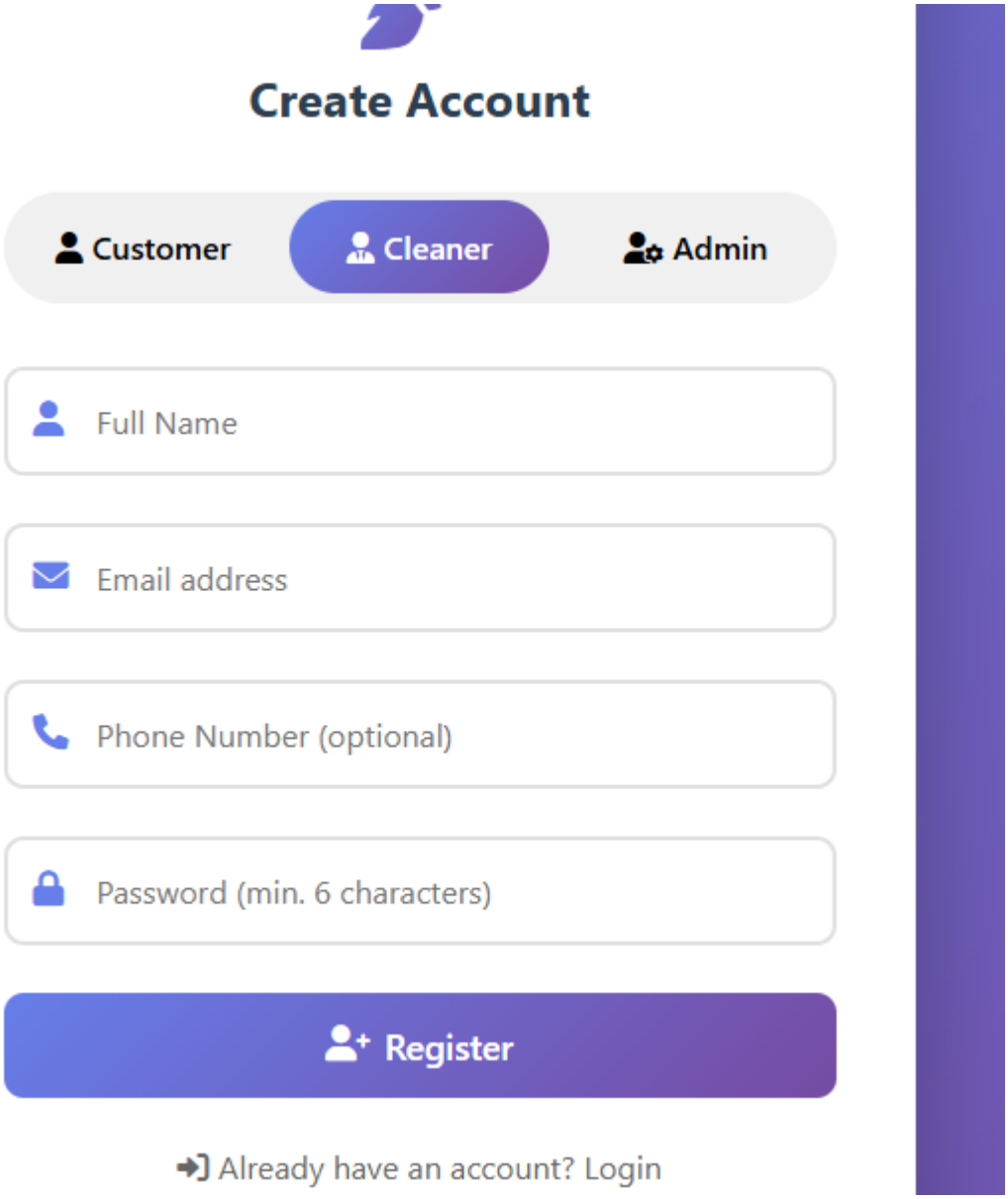
Database Implementation – MySQL stores application data and connects with the backend.

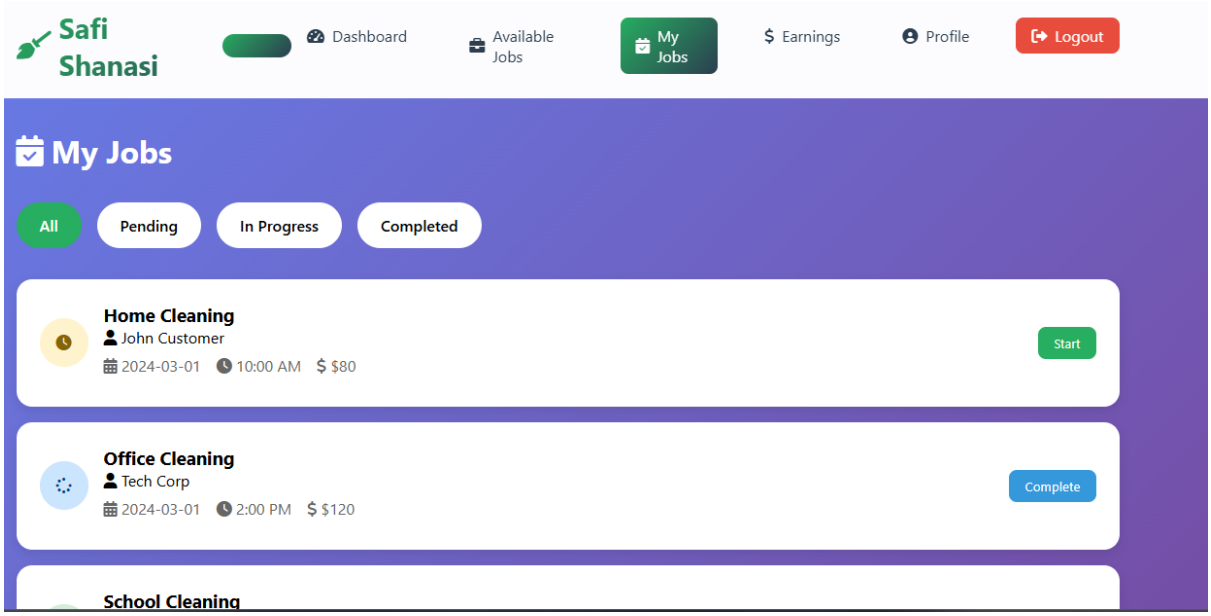
**8. Testing and Results**
The system was tested to ensure all major features work correctly including:
• User registration
• User login
• Viewing services
• Booking services
• Database connectivity
• Dashboard functionality

Screenshots of the system are included in this section showing different parts of the system we explained




Create Account
Customer
Cleaner
Admin
Full Name
Email address
Phone Number (optional)
Password (min. 6 characters)
Register
Already have an account? Login


Safi Shanasi
Dashboard
Available Jobs
My Jobs
Earnings
Profile
Logout
My Jobs
All
Pending
In Progress
Completed
Home Cleaning
John Customer
2024-03-01
10:00 AM
$80
Start
Office Cleaning
Tech Corp
2024-03-01
2:00 PM
$120
Complete
School Cleaning

# Our Services

Choose from our wide range of professional cleaning services

All Services | Home Cleaning | Office Cleaning | School Cleaning

Popular

**Home Deep Cleaning**

Complete home cleaning including bedrooms, kitchen, and living areas

$80.00 /per session
4 hours
Book Now

Popular

**Standard Home Cleaning**

Basic cleaning service for your home

$50.00 /per hour
2 hours
Book Now

Popular

**Commercial Office Cleaning**

Deep cleaning for commercial spaces

$200.00 /per session
5 hours
Book Now

Popular

**Office Sanitization**

Popular

**School Classroom Cleaning**

Popular

**University Facilities**

Safi Shanasi Cleaner

Dashboard | Available Jobs | My Jobs | Earnings | Profile | Logout

## Welcome back, vincent kapanga!

Here's your cleaning service overview

| 12 | 5 | 4.8 | $450 |
|---|---|---|---|
| Available Jobs | My Jobs | Rating | Earnings |

## Available Jobs Near You

**Home Deep Cleaning** Urgent
John Customer
2024-03-15 at 10:00 AM
$80

**Office Sanitization**
Tech Corp
2024-03-15 at 2:00 PM
$120

**School Cleaning** Urgent
International School
2024-03-16 at 9:00 AM
$150

phpMyAdmin

Recent Favorites

- New
- information_schema
- mysql
- performance_schema
- phpmyadmin
- safishanasi
- safi_shanasi
  - New
  - bookings
  - cleaners
  - payments
  - reviews
  - services
  - users
- test

Server: 127.0.0.1 » Database: safi_shanasi

Structure | SQL | Search | Query | Export | Import | Operations | Privileges | Routines | Events | Triggers

Filters

Containing the word:

| Table | Action | Rows | Type | Collation | Size | Overhead |
|---|---|---|---|---|---|---|
| bookings | Browse Structure Search Insert Empty Drop | 3 | InnoDB | utf8mb4_general_ci | 112.0 KiB | - |
| cleaners | Browse Structure Search Insert Empty Drop | 10 | InnoDB | utf8mb4_general_ci | 48.0 KiB | - |
| payments | Browse Structure Search Insert Empty Drop | 1 | InnoDB | utf8mb4_general_ci | 32.0 KiB | - |
| reviews | Browse Structure Search Insert Empty Drop | 1 | InnoDB | utf8mb4_general_ci | 64.0 KiB | - |
| services | Browse Structure Search Insert Empty Drop | 6 | InnoDB | utf8mb4_general_ci | 32.0 KiB | - |
| users | Browse Structure Search Insert Empty Drop | 18 | InnoDB | utf8mb4_general_ci | 64.0 KiB | - |
| 6 tables | Sum | 39 | InnoDB | utf8mb4_general_ci | 352.0 KiB | 0 B |

Check all With selected:

Recent Favorites

- New
- information_schema
- mysql
- performance_schema
- phpmyadmin
- safishanasi
- safi_shanasi
  - New
  - bookings
  - cleaners
  - payments
  - reviews
  - services
  - users
- test

Showing rows 0 - 17 (18 total, Query took 0.0018 seconds.)

```
SELECT * FROM `users`
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

+ Options

| | id | fullName | email | password | phone | role | create |
|---|---|---|---|---|---|---|---|
| Edit Copy Delete | 1 | Admin User | admin@safishanasi.com | $2a$10$N.zmJ9CFsDc9F3qQq5XUqO5X5X5X5X5X5X5X5X5X5... | +255712345678 | admin | 2026-0 |
| Edit Copy Delete | 2 | John Customer | john@example.com | $2a$10$N.zmJ9CFsDc9F3qQq5XUqO5X5X5X5X5X5X5X5X5X5... | +255712345679 | customer | 2026-0 |
| Edit Copy Delete | 3 | Mary Cleaner | mary@example.com | $2a$10$N.zmJ9CFsDc9F3qQq5XUqO5X5X5X5X5X5X5X5X5X5... | +255712345680 | cleaner | 2026-0 |
| Edit Copy Delete | 4 | Sarah Customer | sarah@example.com | $2a$10$N.zmJ9CFsDc9F3qQq5XUqO5X5X5X5X5X5X5X5X5X5... | +255712345681 | customer | 2026-0 |
| Edit Copy Delete | 5 | James Cleaner | james@example.com | $2a$10$N.zmJ9CFsDc9F3qQq5XUqO5X5X5X5X5X5X5X5X5X5... | +255712345682 | cleaner | 2026-0 |
| Edit Copy Delete | 6 | yn | juma@gmail.com | $2b$10$n4O/vvAbiMTMtwPgi5AMSeKAjy48w4SYm1vGc/LnUif... | 0765709202 | admin | 2026-0 |
| Edit Copy Delete | 7 | dreen | dayana@gmail.com | $2b$10$76.DESJ6FHcJrZu9U4.AAem.YC2BRfyOYfQaphK2Fs3... | 07121311244 | cleaner | 2026-0 |
| Edit Copy Delete | 8 | dreen | dayaa@gmail.com | $2b$10$Nn4As1ctLR4CInalcsvlM.u196DSjIpHSKsYAKxJsm3... | 07121311244 | cleaner | 2026-0 |
| Edit Copy Delete | 9 | ujytf | johhn@example.com | $2b$10$YCYRxthD/K7F3kP/xDkTS.iZn6mgvXaom7cJ84XA2CM... | 0765709202 | customer | 2026-0 |

Console

## 9. Conclusion

The SafishaNasi App successfully provides an online platform for booking cleaning services. The system integrates frontend, backend, and database technologies to create a functional full stack web application.

## 10. Future Improvements

Possible improvements include:

- Integration with mobile payment systems such as M-Pesa
- Cleaner rating and review system
- Real-time notifications
- Mobile application version
- Advanced analytics dashboard for administrators